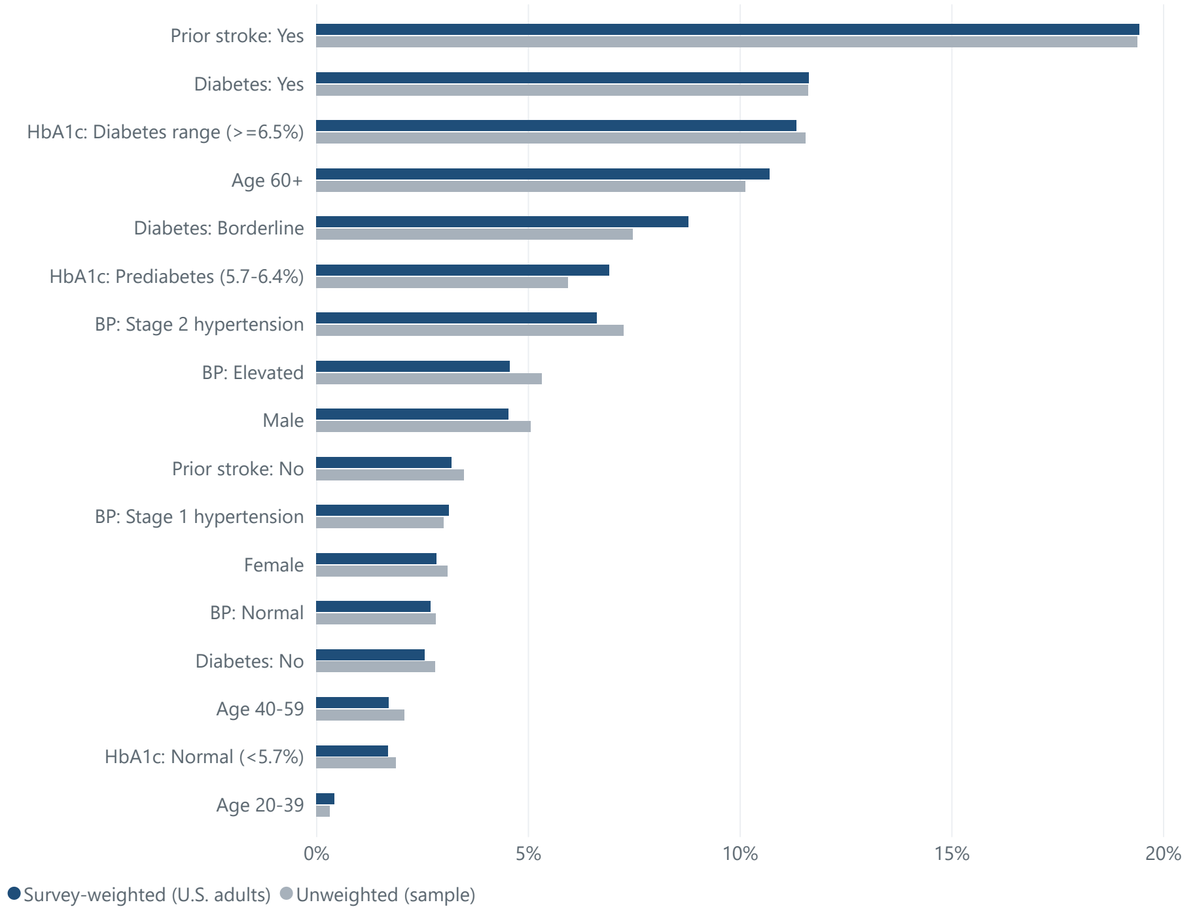
Coronary heart disease in U.S. adults

Survey-weighted NHANES estimates, 2013 to March 2020 (three cycles, 20,469 adults aged 20+ examined). CHD = self-reported doctor diagnosis.

U.S. adult CHD prevalence, 2013-2020	95% CI (design-based)	CHD cases (3 cycles)	Adults examined
3.8%	3.3% to 4.5%	891	20,469

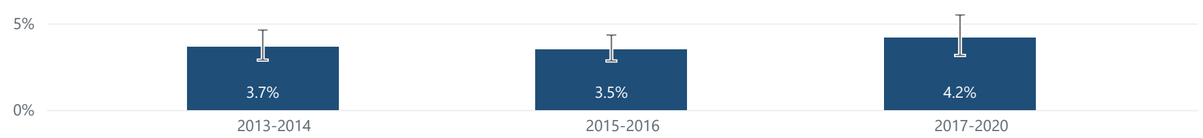
CHD prevalence by risk group, 2013-2014: survey-weighted vs unweighted

Weighted = U.S. adult population estimate (exam weights); unweighted = the sample. Highest after a stroke, with diabetes and at 60+.



CHD prevalence (survey-weighted) by NHANES cycle

Pooled 2013-March 2020: 3.8% (95% CI 3.3-4.5%). No clear change across cycles; CIs overlap.



Adjusted odds ratios for CHD, U.S. adults 2013-2020 (survey-weighted, 95% CI)

Navy square = 95% CI excludes 1; grey = includes 1. Per SD for continuous measures; adjusted for all predictors and survey cycle.

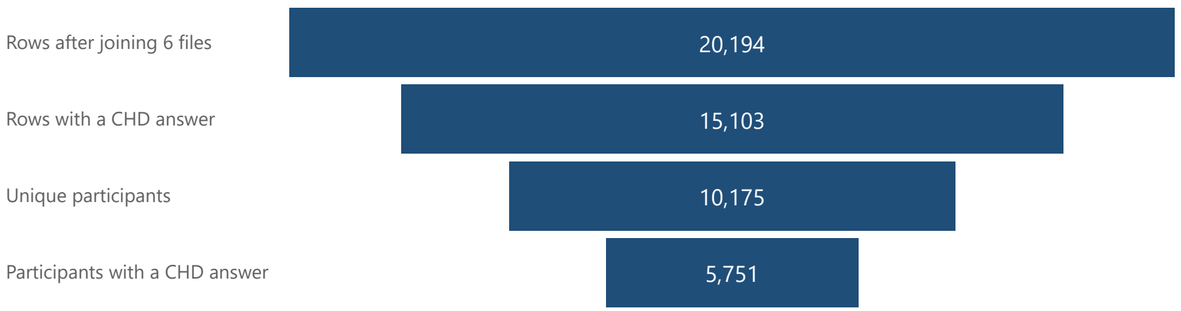
Data audit: why the original AUC of 0.999 was wrong

The medications file has one row per prescription. Joining it on participant ID repeated people up to 23 times, so copies of the same person sat in both training and test sets.

Rows per participant in the original data	Test rows with a copy in training	Random forest AUC, original (leaky)	Random forest AUC, corrected
2.63	75.4%	0.999	0.858

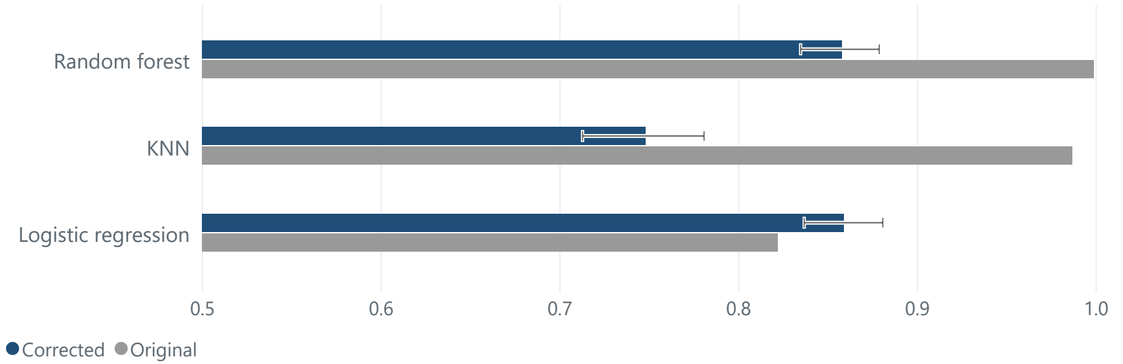
Rows vs people: the medications join multiplied participants

15,103 labeled rows were only 5,751 people (2.63 rows each); copies landed in both train and test.



AUC before vs after the audit: grey = original row-level split, blue = participant-level CV

Random forest 0.999 to 0.858, KNN 0.987 to 0.748, logistic regression 0.822 to 0.859. Whiskers = bootstrap 95% CI; axis starts at 0.5 (chance).



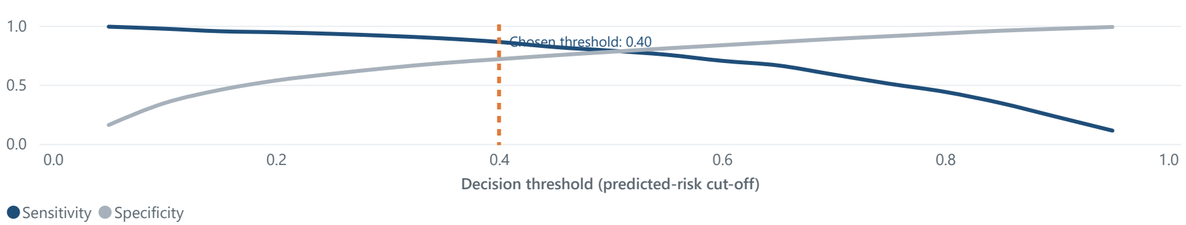
Decision threshold (drag to change)

0.40

Sensitivity	Specificity	Adults flagged
86%	72%	31%

Choosing a screening threshold (logistic regression, participant-level CV)

Drag the threshold slider above: the orange line and the three cards update. At 0.40 the model catches 86% of CHD cases at 72% specificity.



What the corrected pipeline does differently

- Collapses medications to one row per participant before joining, and checks that every ID is unique.
- Validates with participant-level stratified 5-fold cross-validation; imputation and scaling are fit inside each fold.
- Reports AUC with bootstrap 95% CIs instead of accuracy, which is misleading at 4% prevalence.
- Picks the decision threshold for screening sensitivity (0.40) instead of the 0.5 default.
- Relabels MCQ160F as the participant's own stroke history; it had been labeled as a blood relative's.

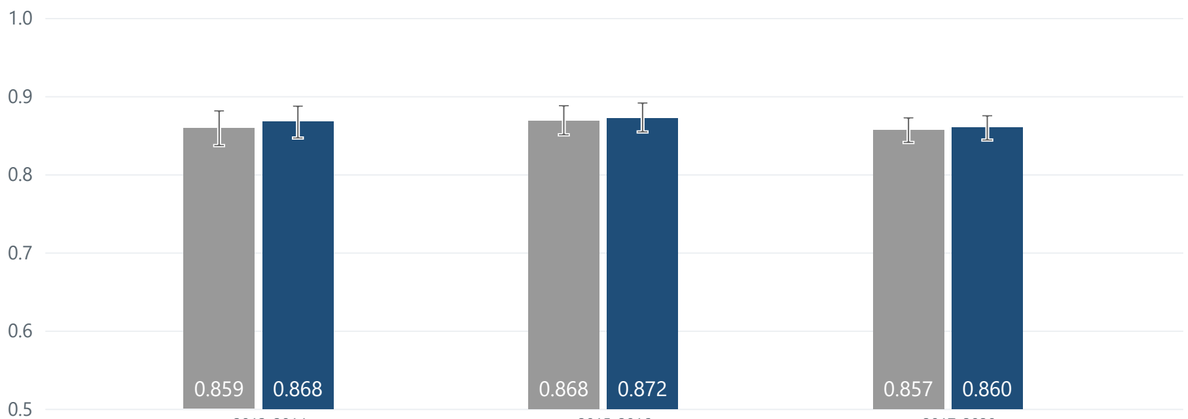
Result: random forest drops from 0.999 to 0.858, and plain logistic regression (0.859) does just as well.

Model validation: does the model hold up in later NHANES cycles?

Temporal validation within NHANES: logistic regression with the 19 predictors, trained on 2013-2014 and applied unchanged to 2015-2016 and 2017-March 2020, plus leave-one-cycle-out cross-validation.

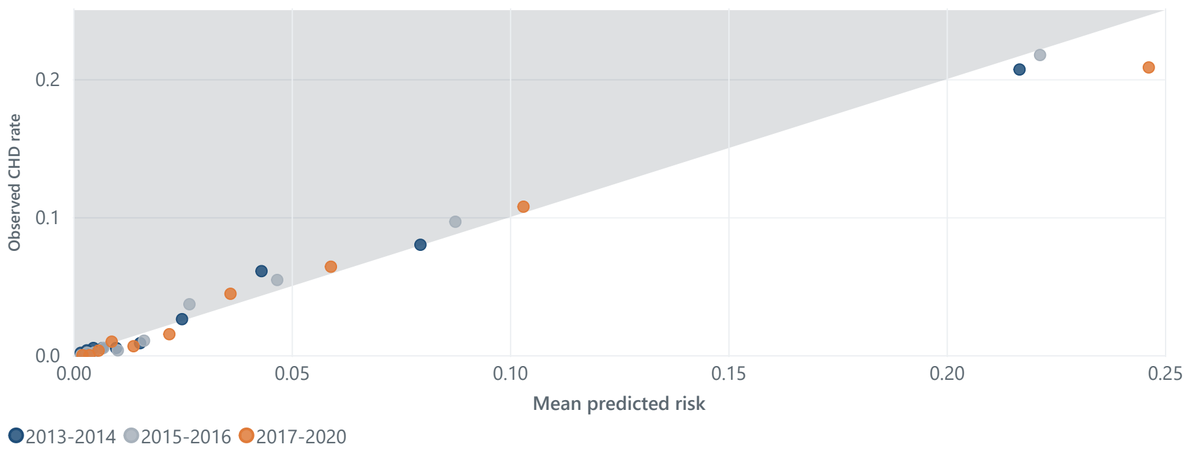
Discrimination holds in every NHANES cycle (AUC, logistic regression)

Grey: trained on 2013-2014 only (its 2013-2014 bar is internal CV). Blue: leave-one-cycle-out. Whiskers = bootstrap 95% CI; axis starts at 0.5 (chance).



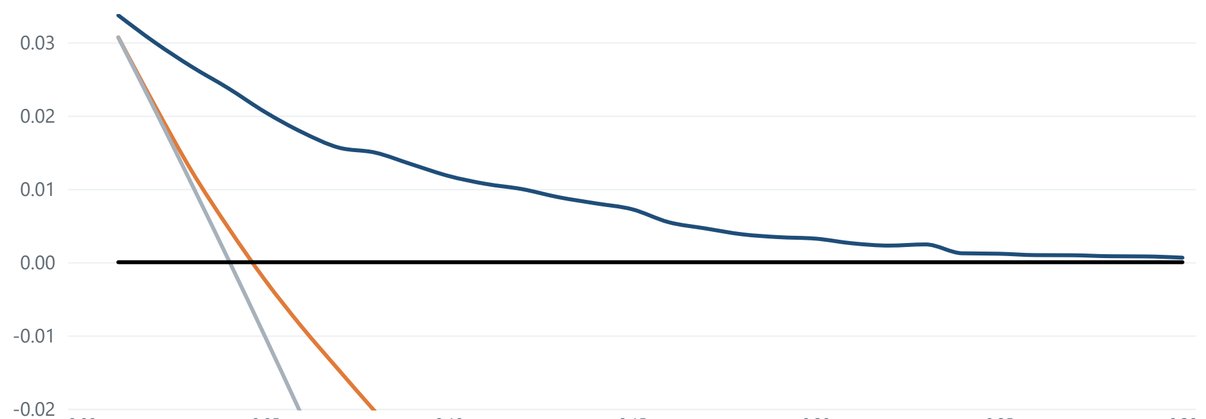
Calibration: predicted vs observed CHD risk by decile (unweighted logistic regression)

Each dot is one risk decile; the edge of the grey area is perfect calibration (observed = predicted). Trained on 2013-2014, applied unchanged. Observed/expected 0.99, 1.01, 0.92; calibration slope 0.94-0.99.



Decision curve: net benefit across treatment thresholds (2013-2014, out-of-fold)

The calibrated (unweighted) model beats treat-all and treat-none at every threshold from 1% to 30%; the class-weighted model falls below treat-none above 5%.



What the validation shows

- Trained once on 2013-2014 and applied unchanged to 2015-2016 and 2017-March 2020: AUC 0.868 and 0.857 (internal CV 0.859).
- Leave-one-cycle-out (each cycle held out once): AUC 0.86-0.87 in every cycle, calibration slope 0.98-1.00.
- More training data raises AUC only slightly (+0.003 to +0.009): the 19 predictors, not sample size, limit discrimination.
- Robust to the 2017 switch to an oscillometric blood-pressure device: dropping BP changes external AUC by < 0.001.
- Use the unweighted model for risk estimates; class weighting inflates predicted risk about sevenfold.